DATA COLLECTION AND TRENDING STEVENAGE
QUOTE REFERENCE NO. 250728

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1 Revision History

REVISION	DETAILS	AUTHOR	DATE
0.0	Initial Issue	Harrison Wardle	14/07/2025
1.0	Internal Review	John Pirie-Elsom	24/07/2025
2.0	Client Issue		
2.1	Comments Reviewed	Dean Holland	07/08/2025
2.2	Comments Reviewed	Dean Holland	20/08/2025
2.3	Network Drawing And Layout Updated	Dean Holland	02/09/2025
2.4	Network Drawing Updated, 3.2.3 & 3.2.4 Removed	Dean Holland	09/09/2025
2.5	Comments Reviewed	Dean Holland	22/09/2025
2.6	Comments Reviewed - Units Updated	Dean Holland	22/09/2025
2.7	Flow Rate Units Removed	Dean Holland	23/09/2025
2.8	Enclosure Information	Dean Holland	23/09/2025
3.0	Client Acceptance		

2 Document Approval

Customer Ref No: **Data Collection and Trending**

	Name	Signature	Date
Document Author	Harrison Wardle	H. Wardle	14/07/2025
Document Author	John Naranjo		30/09/2025
Document Author	Francois Pierrel		30/09/2025
Checked By	John Pirie-Elsom	J Pirie-Elsom	24/07/2025
Client Approval			

AJS Project No: 250728

Document Title: Functional Design Specification

Site Location: Europe Snacks, Stevenage

Signature of the document below constitutes acceptance of the latest issue shown. The accepted document provides the basis for all the work undertaken.

The Electrical Installation which may consist of the following:
Site Installation
Control Panels

Will be Designed, Built and Installed in conjunction with standards set out in the following document.

AJS Control and Installation Standards R5

The above document which is available on request has been written using BS 60204-1:2018 Safety of Machinery as reference.

3 Project Information

3.1 Project Summary

This Functional Design Specification (FDS) outlines the design and operational requirements for the Data Collection & Trending system to be implemented at Europe Snacks' Stevenage facility. The system is designed to capture electrical and gas energy usage data from frying and ancillary lines, integrating with the factory's network for real-time data display on a virtual machine (VM). A Human-Machine Interface (HMI) will be installed on the main control panel to display data and provide system optimization options. The system will collect data from existing and new gas and electric meters, with pulse signals used to process gas and electric meter data into Nm3 and kW values.

The project includes the design, supply, installation, and commissioning of control panels, electrical and gas metering systems, and associated software. The scope covers the specific production lines listed below, as detailed in the proposal (CA-250728-5).

- Pub Bacon
- Pub Scampi (Fryer)
- Pub Scampi (Dryer)
- Stackers A
- Stackers B
- Cheetos (Planet Dryer)
- Stackers Compressor (Compressor 1)
- Compressor 2
- HVAC (Stackers B)

3.1.1 Objectives:

- Collect and monitor energy usage from specified production lines.
  - Electricity in kW
  - Gas in Nm^3
- Provide real-time data visualization on a Siemens Unified Comfort HMI and client VM.
- Ensure reliable integration with the site's network infrastructure.
- Comply with IEC standards and safety requirements (BS EN 60204-1).

3.2 Project Scope

AJS are planning to design and procure the following:

3.2.1 3.2.1 ENC01 (Main HMI Panel)

Located in the main switch room

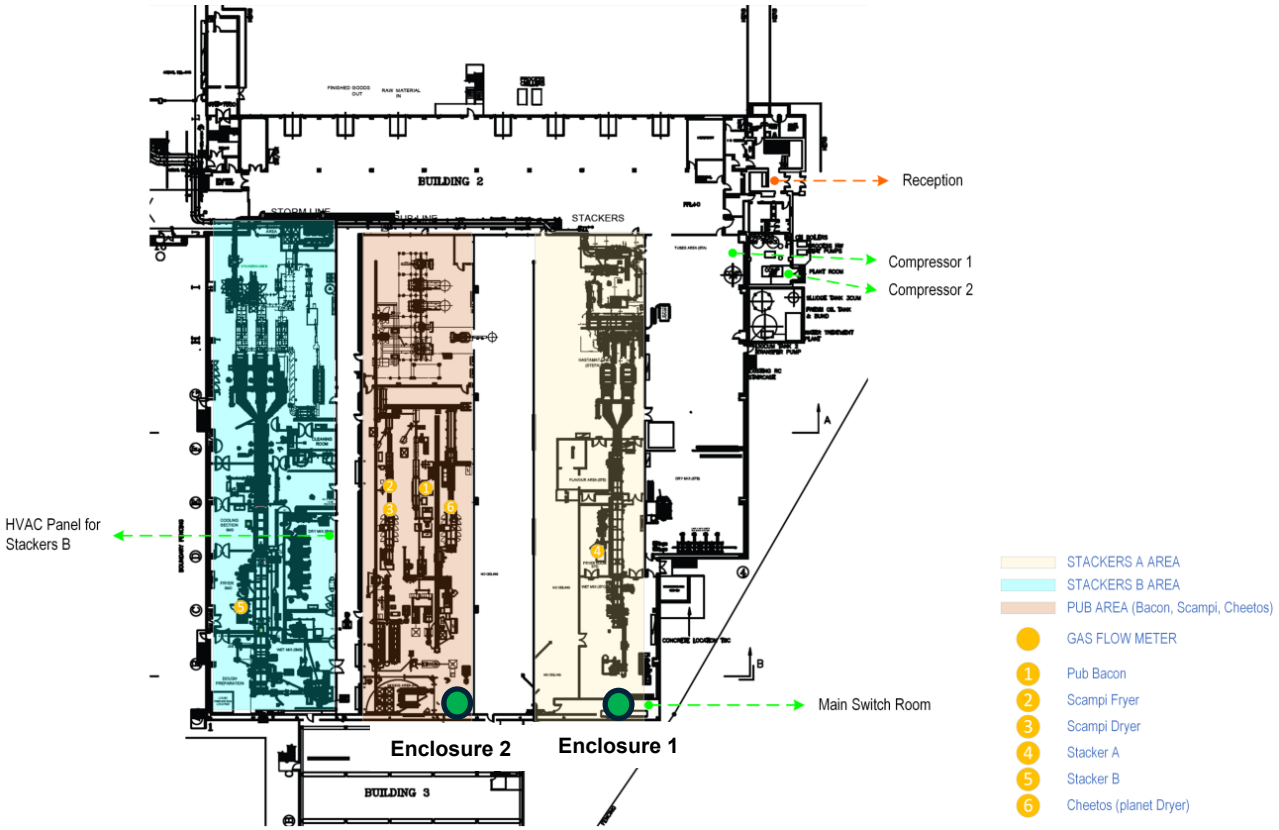
- Mild steel Rittal enclosure
- Main Switch
- Commissioning Socket
- Siemens Unified Comfort HMI 10"
- Siemens ET200SP 1500CPU
- Siemens HF Digital Input Cards x2
- Network Security Switch

3.2.2 3.2.2 ENC02 (Pub)

Located in the pub area mains cupboard

- Mild steel Rittal enclosure
- Main Switch
- Commissioning Socket
- Siemens ET200SP Interface
- Siemens HF Digital Input Cards x2
- Network Switch

3.3 Project Drawing Overview



4 Control Panel

4.1 Devices

4.1.1 Enclosure Door Furniture

- Simatic Unified HMI (6AV21283KB060AX1)
- Socomec Isolator Handle (Blue and Black) – 22004016 (Main Isolator)

4.1.2 Low Voltage

- 8port B008 Scalance Network Switch
- Socomec Main Isolator? A interlocked with Panel Door
- Phoenix Contact Programming Socket.
- Siemens RCBO, 6A, 30mA.
- ?A Sitop 24VDC PSU 230v.

4.1.3 Extra Low Voltage

- 3 x Electronic Miniature Circuit Breakers
- CPU ET200 1510SP F-1 PN Processor
- SIMATIC HMI MTP1000 Unified Comfort Panel
- ? Port Managed Ethernet Switch
- System Healthy Indicator Lamp?

4.1.4 Network

- ENC01 CPU ET200 1500 Processor
- ? Port Managed Ethernet Switch
- SIMATIC HMI MTP1000 Unified Comfort Panel
- ENC02 ET200 Interface

4.1.5 Main PLC

PLC CPU will be configured to attempt warm restart to RUN after any power interruption. The new control panel will consist of the following devices.

- CPU ET200 1500 Processor
- HF Digital Input Cards x4

4.2 Control Philosophy

4.2.1 Operator control

The system will use an HMI to visually display data and trending on the panel. Operators can see trending graph and data real time data coming into the PLC. Alarms and warnings will indicate to the operator if the data isn't getting collected from the specified Meters.

4.2.2 Data Acquisition

Collect pulse signals from electric and gas meters to monitor energy usage across all specified lines.

4.2.3 Data Processing

Convert pulse inputs into usable metrics for trending and analysis.

- Electricity in kW
- Gas Meters in Nm3

4.2.4 Visualization

Display real-time and historical energy usage data on a Siemens Unified Comfort HMI and client virtual machine (VM).

4.2.5 Functions

Data will be collected and then calculations will be executed within the PLC, calculated values will be sent to the client virtual machine (VM), where that data will be stored
For each line:

- Gas Meters
  • Pub Bacon • Pub Scampi (Fryer) • Pub Scampi (Dryer) • Stackers A • Stackers B • Cheetos (Planet Dryer)
  - Totaliser every 10 mins Units: Nm^3
- Electric Meters:
  • Stackers Compressor (Compressor 1) • Compressor 2 • HVAC (Stackers B)
  - Totaliser every 10 mins Units: kW

4.2.6 Data Store

We will only hold limited amounts of data in the event of an error or loss communication between AJS and Enertherm VM

5 Software

5.1.1 Platform

The software will be created on TIA Portal V19 as per the customers' requirements and will follow any programming guidelines they provide.

5.1.2 Development

PLC code and HMI graphics will be developed to collect, process, and display data as per the scope.

5.1.3 Testing

Off-site simulation and Factory Acceptance Test (FAT) to verify functionality per this FDS.

5.1.4 Commissioning

On-site commissioning to ensure system operation aligns with the FDS, including I/O testing and system optimization.

5.1.5 Licences

No additional software licenses are required.

5.2 PLC

PLC CPU will be configured to attempt warm restart to RUN after any power interruption.

5.3 Alarms (To be confirmed)

Alarm Text on HMI	Description
<MeterTag> – Communication Lost	No new data from meter for > **60 s** (or 5× scan period).
<MeterTag> – Value Out of Sensor Range	Reported value below/above engineered min/max (e.g., kW, Nm³).

5.4 HMI (To be Confirmed)

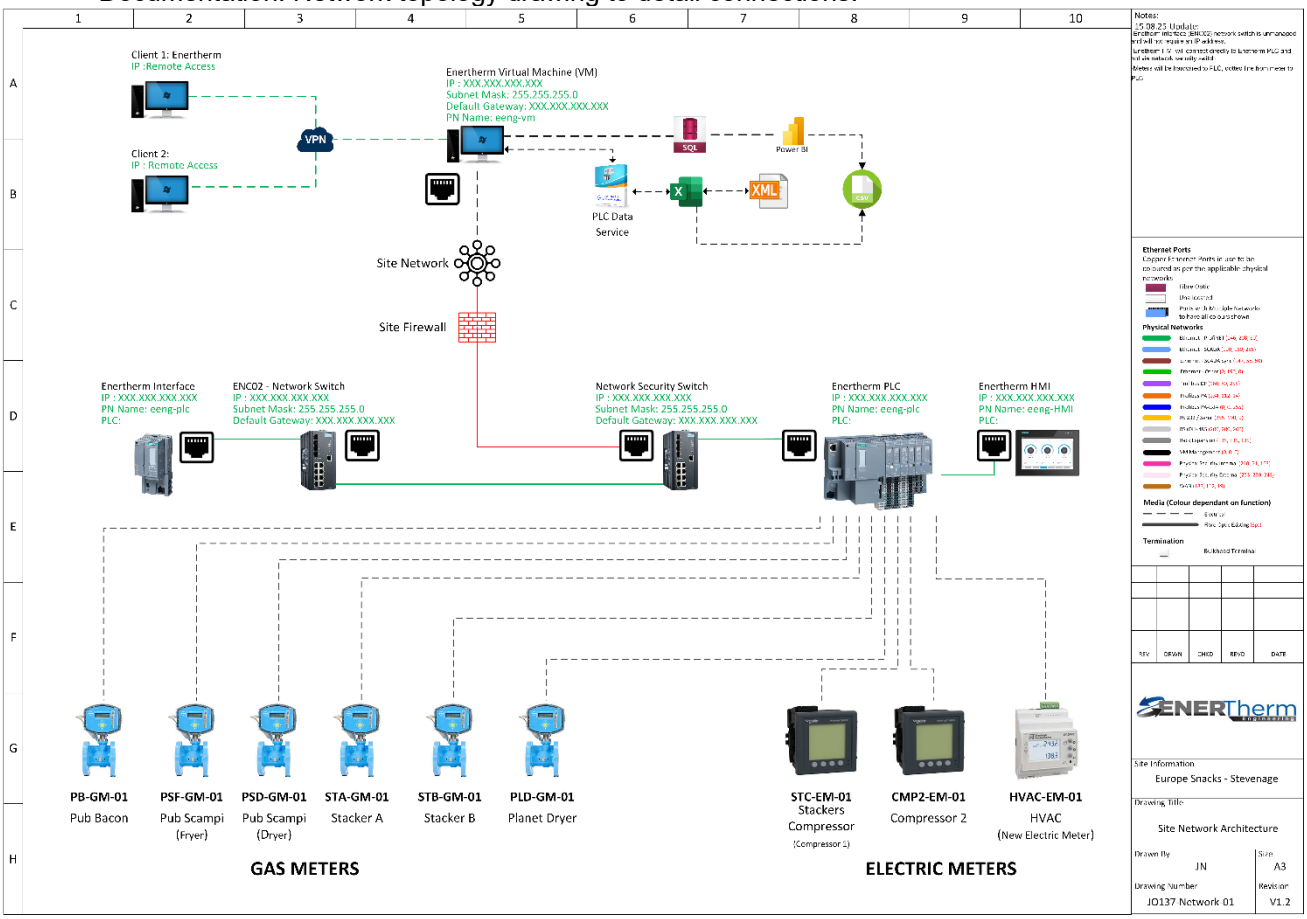
The HMI will consist of the additional screens below –

Screen Number	Screen Name	Description
01	**Home Page**	System overview and navigation.
02	**Overview Page**	Summary of energy usage across all lines.
03	**Engineering Settings**	Configuration options for system parameters.
04	**Alarms**	Alerts for system anomalies or meter faults.
05	**Trending Screens**	Graphical representation of current energy usage and historical.: <u>Stevenage Gas Meter:</u> • Pub Bacon • Pub Scampi (Fryer) • Pub Scampi (Dryer) • Stackers A • Stackers B • Cheetos (Planet Dryer) <u>Stevenage Electric Meters</u> • Stackers Compressor (Compressor 1) • Compressor 2 • HVAC (Stackers B)
06	**Service Page**	Maintenance and diagnostic tools.
07	**Constants Page**	Authorized users to change Constant values that are required.

5.5 Network

The network type will be a star network with each network component connected to its specified port on the switch

- Type: Dedicated network cabling, segregated from mains and controls wiring.
- Connection: Via local data cabinet, using a dedicated port on the site's patch panel.
- Documentation: Network topology drawing to detail connections.



6 Safety

6.1 Safety Requirements

There will be no Safety Requirements for this project

7 Assumptions and Notes

- All data collection is based on pulse signals from new and existing meters.
- The site will provide a dedicated network port for integration.

8 Field Devices

8.1.1 Electric Meters:

- Quantity: 1 new Rayleigh Connect RI-D440 meters for pulse wiring.
- New: 1 Meters
- Old: 2 Meters
- Monitored Lines • Stackers Compressor (Compressor 1) • Compressor 2 • HVAC (Stackers B)

- Output: Pulse signals for kW measurement.

8.1.2 Gas Meters:

- Quantity: 6 new flow meters for pulse wiring.
- New: 6 Meters
- Old: N/A
- Monitored Lines • Pub Bacon • Pub Scampi (Fryer) • Pub Scampi (Dryer) • Stackers A • Stackers B • Cheetos (Planet Dryer)
- Output: - Pulse signals for Nm³ measurement

8.1.3 Tags:

Gas Meters:

- Pub Bacon: PB-GM-01
- Pub Scampi (Fryer): PSF-GM-01
- Pub Scampi (Dryer): PSD-GM-01
- Stacker A: STA-GM-01
- Stacker B: STB-GM-01
- Planet Dryer (Cheetos): PLD-GM-01

Electric Meter:

- Stackers Compressor: STC-EM-01
- Compressor 2: CMP2-EM-01
- HVAC: HVAC-EM-01

8.2 Cable

Network – CAT6 Screened / PROFINET
Control – BS6500 CY/YY Cable
Panel Mains – SWA
Equipotential Bonding – GR/YE BS 6004 – 10mm

- I/O testing of controls wiring and testing of network cabling to carried out and recorded.
- Each cable will be labelled with a printed tag to show cable number.
- Any required junction boxes will be labelled with a printed self-adhesive marker.

9 Conformity

This project will be designed, built and installed in conjunction with standards set out in the AJS Control and Installation Standards R5, in accordance with IEC Standards and meet the requirements of 60204-1:2018 Safety of Machinery - Electrical Equipment of Machines. Furthermore, specific UK Standards will be followed depending on the requirements of the project.

On completion, the control panel(s) will be assessed in accordance with the relevant UK Regulations and UK Designated Standards.

9.1 Declaration of Conformity

A Declaration of Conformity to these standards will be produced for the AJS Control Panel(s) only. This will not extend to new or existing site-based machinery unless specifically referenced in the AJS quotation document.

9.2 UKCA

A physical declaration of compliance will be placed onto the AJS Control Panel(s) only in the form of a UKCA mark. It is assumed that the AJS manufactured product is to be placed on the UK market (client based in the UK). The UKCA mark will not extend to new or existing site-based machinery or systems unless specifically referenced in the AJS quotation document. Re-certification of existing site equipment and systems will be a chargeable additional cost to this project.

9.3 PUWER

It is the client's responsibility to complete a PUWER assessment of all work equipment, new or existing. Unless specifically referenced in the AJS quotation document for this project, the PUWER assessment is excluded from the scope of works.